MATH4041/MCSC6120 Assignment 1

Glucose Control, Part 1

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1. Derivation of the Two-Degree-of-Freedom Model

We begin with the glucose control model introduced in lecture, which consists of an integral equation and a differential equation:

$$u = \alpha v + \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\, dt', \tag{1}$$

and

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t). \tag{2}$$

Our goal is to eliminate the integral so that the model can be expressed as a system of two ordinary differential equations. To do this, we first differentiate the equation for u with respect to time:

$$\dot{u} = \alpha \dot{v} + \beta\lambda \frac{d}{dt}\left(\int_{-\infty}^{t} e^{-\lambda(t-t')} v(t')\, dt' \right). \tag{3}$$

Since the upper limit of integration depends on t, we apply Leibniz's rule:

$$\dot{u} = \alpha \dot{v} + \beta\lambda \left(v(t) - \lambda \int_{-\infty}^{t} e^{-\lambda(t-t')} v(t')\, dt' \right) \tag{4}$$

$$= \alpha \dot{v} + \beta\lambda v - \lambda \left(\beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\, dt' \right). \tag{5}$$

Notice that the remaining integral is the same one that appears in our original equation for u. Rearranging that equation gives:

$$\beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\, dt' = u - \alpha v. \tag{6}$$

We can now substitute this expression into our derivative:

$$\dot{u} = \alpha \dot{v} + \beta\lambda v - \lambda(u - \alpha v) \tag{7}$$

$$= \alpha \dot{v} + \lambda(\alpha + \beta) v - \lambda u. \tag{8}$$

Finally, we introduce the notation:

$$\dot{v} = f(u, v, t) = -\gamma v - \delta u(v + \bar{v}) + F(t). \tag{9}$$

This allows us to express the original model entirely in terms of two differential equations:

$$\boxed{\begin{aligned} \dot{u} &= -\lambda u + \lambda(\alpha + \beta) v + \alpha f(u, v, t), \\ \dot{v} &= -\gamma v - \delta u(v + \bar{v}) + F(t). \end{aligned}} \tag{10}$$

We have therefore obtained the required two-degree-of-freedom ODE system without the integral term.

2. Scaling the Variables

We now want to simplify the system by introducing dimensionless variables, reducing the number of parameters appearing in the equations.

Recall the system obtained in Question 1:

$$\begin{aligned} \dot{u} &= -\lambda u + \lambda(\alpha+\beta)v + \alpha f(u,v,t), \\ \dot{v} &= -\gamma v - \delta u(v+\bar{v}) + F(t). \end{aligned} \tag{11}$$

Following the scaling approach from lecture, we introduce:

$$t = \tilde{t}\,t', \qquad u = \tilde{u}\,u', \qquad v = \bar{v}\,v'. \tag{12}$$

Because time is also being rescaled, we must account for this when transforming the derivatives:

$$\dot{u} = \frac{\tilde{u}}{\tilde{t}}\frac{du'}{dt'}, \qquad \dot{v} = \frac{\bar{v}}{\tilde{t}}\frac{dv'}{dt'}. \tag{13}$$

We also define the dimensionless function f' by:

$$f'(u',v',t') = \frac{\tilde{t}}{\bar{v}} f(\tilde{u}u', \bar{v}v', \tilde{t}t'). \tag{14}$$

Substituting the new variables into both equations gives:

$$\frac{\tilde{u}}{\tilde{t}}\frac{du'}{dt'} = -\lambda\tilde{u}u' + \lambda(\alpha+\beta)\bar{v}v' + \alpha\frac{\bar{v}}{\tilde{t}}f'(u',v',t'), \tag{15}$$

$$\frac{\bar{v}}{\tilde{t}}\frac{dv'}{dt'} = -\gamma\bar{v}v' - \delta\tilde{u}u'(\bar{v}v'+\bar{v}) + F(\tilde{t}t'). \tag{16}$$

Next, we isolate the derivatives by dividing through by their respective coefficients:

$$\frac{du'}{dt'} = -(\lambda\tilde{t})u' + \frac{\lambda\bar{v}\tilde{t}}{\tilde{u}}(\alpha+\beta)v' + \frac{\alpha\bar{v}}{\tilde{u}}f'(u',v',t'), \tag{17}$$

$$\frac{dv'}{dt'} = -(\gamma\tilde{t})v' - (\delta\tilde{u}\tilde{t})u'(v'+1) + \frac{\tilde{t}}{\bar{v}}F(\tilde{t}t'). \tag{18}$$

To make the second equation simpler, we choose the time and u-scales such that:

$$\gamma\tilde{t} = 1, \qquad \delta\tilde{u}\tilde{t} = 1. \tag{19}$$

This gives:

$$\boxed{\tilde{t} = \frac{1}{\gamma}, \qquad \tilde{u} = \frac{1}{\delta\tilde{t}} = \frac{\gamma}{\delta}.} \tag{20}$$

We also introduce the scaled input:

$$\boxed{G(t') = \frac{F(\tilde{t}t')}{\gamma\bar{v}}.} \tag{21}$$

With these choices, the system becomes:

$$\frac{du'}{dt'} = -\frac{\lambda}{\gamma}u' + \frac{\lambda\delta\bar{v}}{\gamma^2}(\alpha+\beta)v' + \frac{\alpha\delta\bar{v}}{\gamma}f'(u',v',t'), \tag{22}$$

$$\frac{dv'}{dt'} = -v' - u'(v'+1) + G(t'). \tag{23}$$

We can simplify the notation further by defining three new parameters:

$$\boxed{a = \frac{\lambda}{\gamma}, \qquad b = \frac{\lambda\delta\bar{v}}{\gamma^2}(\alpha+\beta), \qquad c = \frac{\alpha\delta\bar{v}}{\gamma}.} \tag{24}$$

Using these definitions, the dimensionless system can be written more compactly as:

$$\boxed{\begin{aligned} \frac{du'}{dt'} &= -au' + bv' + cf'(u',v',t'), \\ \frac{dv'}{dt'} &= f'(u',v',t') = -v' - u'(v'+1) + G(t'). \end{aligned}} \tag{25}$$

For the equilibrium calculations and numerical experiments, we take G to be constant. This completes the scaling process, leaving us with the reduced parameters a, b, c, and G.

3. Equilibrium Solutions

To determine the equilibrium solutions, we set both time derivatives equal to zero. Since our equations are now dimensionless, we can drop the primes for simplicity.

The equilibrium conditions are:

$$\boxed{\begin{aligned} -au + bv &= 0, \\ -v - u(v+1) + G &= 0. \end{aligned}} \tag{26}$$

At equilibrium, $f(u,v) = 0$ because it represents the v-equation, whose derivative must also vanish.

We begin by solving the first equation for u:

$$-au + bv = 0. \tag{27}$$

Rearranging gives:

$$\boxed{u^* = \frac{b}{a}v^*.} \tag{28}$$

We then substitute this relationship into the second equilibrium equation:

$$-v - \frac{b}{a}v(v+1) + G = 0. \tag{29}$$

Expanding and collecting terms:

$$-\frac{b}{a}v^2 - \frac{b}{a}v - v + G = 0, \tag{30}$$

$$v^2 + \left(1 + \frac{a}{b}\right)v - \frac{a}{b}G = 0. \tag{31}$$

This is a quadratic equation in v, so applying the quadratic formula gives:

$$\boxed{v^* = -\frac{1}{2}\left(1 + \frac{a}{b}\right) \pm \frac{1}{2}\sqrt{\left(1 + \frac{a}{b}\right)^2 + \frac{4a}{b}G.}} \tag{32}$$

The number of real equilibrium solutions depends on the expression inside the square root. In particular, the two solutions coincide when:

$$\left(1 + \frac{a}{b}\right)^2 + \frac{4a}{b}G = 0. \tag{33}$$

Solving this equation for G gives the critical saddle-node value:

$$\boxed{G_{\text{SN}} = -\frac{b}{4a}\left(1 + \frac{a}{b}\right)^2.} \tag{34}$$

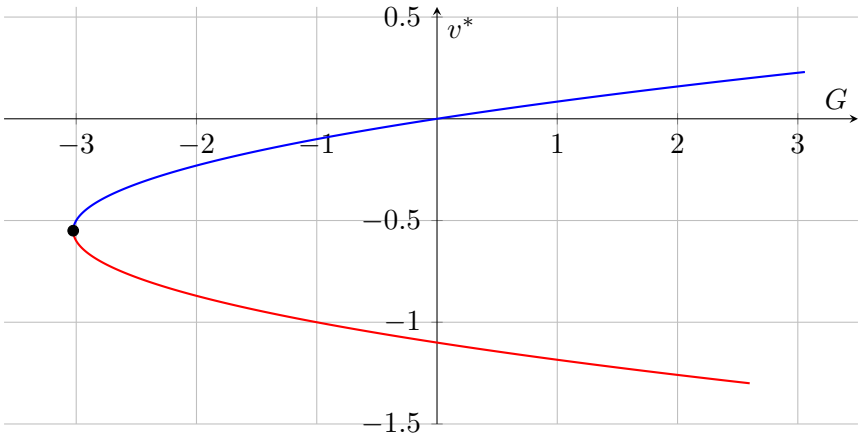
Assuming $a > 0$ and $b > 0$, this critical value separates the three possible cases. For $G < G_{\text{SN}}$, there are no real equilibria. At $G = G_{\text{SN}}$, the two solutions merge into one, while for $G > G_{\text{SN}}$, two distinct real equilibria exist.

We can summarize these results in the following table:

Equilibrium Solutions as a Result of G		
Value of G	**Number of Solutions**	**Equilibrium Solution(s)**
$G < G_{\text{SN}}$	Zero	N/a
$G = G_{\text{SN}}$	One	$(u^*, v^*) = \left(\frac{b}{a}v^*, -\frac{1}{2}\left(1 + \frac{a}{b}\right)\right)$
$G > G_{\text{SN}}$	Two	$(u^*, v^*) = \left(\frac{b}{a}v^*, -\frac{1}{2}\left(1 + \frac{a}{b}\right) \pm \frac{1}{2}\sqrt{\left(1 + \frac{a}{b}\right)^2 + \frac{4a}{b}G}\right)$

To visualize these results, we can sketch the equilibrium branch against the input parameter G. Using the same values as the reference example, $a = 0.1$ and $b = 1.0$, we can rearrange the equilibrium equation as:

$$G = v^* + \frac{b}{a}v^*(v^* + 1) = 10(v^*)^2 + 11v^*. \tag{35}$$



As shown in the sketch, the two branches meet at $G_{\text{SN}} = -3.025$. Below this value, no real equilibrium solutions exist. Above it, we obtain two separate equilibrium solutions.

4. Numerical Tests of Controllability

To investigate the controllability of the non-dimensional system, numerical experiments were conducted using the supplied Python script `g_simul.py`. Following Question 3, parameter values $a = 0.1$ and $b = 1.0$ were held fixed, giving a saddle-node threshold of:

$$G_{\text{SN}} = -\frac{1}{4(0.1)}(1.1)^2 = -3.025. \tag{36}$$

Trajectories were simulated using a second-order Runge-Kutta scheme across representative values of G selected from each of the three parameter regimes identified in Question 3:

1. **Case 1 ($G < G_{\text{SN}}$):** $G = -4.0$ (no equilibrium points).
2. **Case 2 ($G = G_{\text{SN}}$):** $G = -3.025$ (single saddle point).
3. **Case 3 ($G > G_{\text{SN}}$):** $G = 1.6$ (two equilibrium points).

A trajectory was classified as **controlled** if it converged to a stable equilibrium point, and **uncontrolled** if it diverged toward $v \to -\infty$ (signaled by $v < -100$).

Case 1: $G = -3.7 < G_{\text{SN}}$

Here, the discriminant from Question 3 is negative, so there are **no real equilibria**. I tried four initial conditions:

$$(12, 0), \qquad (0, 12), \qquad (-12, 0), \qquad (0, -12).$$

Every run eventually reached the negative-v cutoff. This is consistent with the equilibrium calculation: there is no steady state for these solutions to approach.

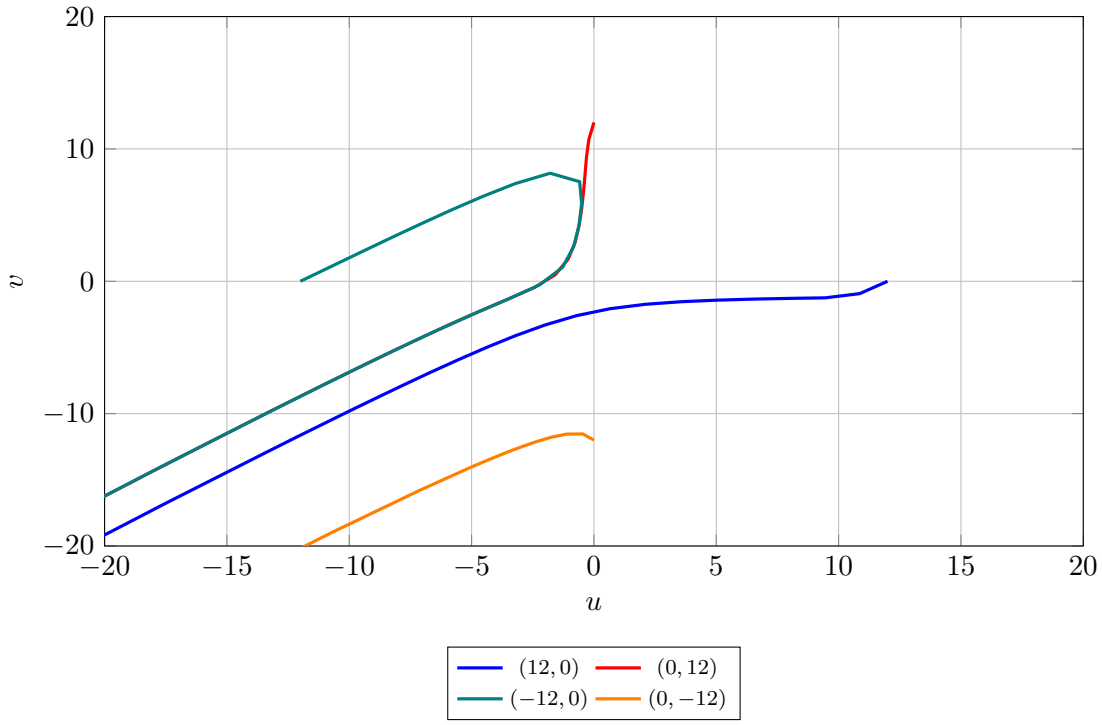


Figure 1: Trajectories for $G = -3.7$ from the four tested starting points.

Case 2: $G = -3.025 = G_{\text{SN}}$

At the critical value, the two equilibrium branches meet. Using $u^* = 10v^*$, the only equilibrium is

$$(u^*, v^*) = (-5.5, -0.55).$$

The Jacobian has eigenvalues 0 and 3.95. Thus, this is a **nonhyperbolic equilibrium**, with an unstable direction corresponding to the positive eigenvalue.

For this case, my initial conditions were

$$(-6, -1.3), \quad (-5, 0), \quad (-4.5, 0.2), \quad (-7, -0.8).$$

All four trajectories eventually passed the cutoff instead of remaining near the equilibrium. This only describes the points I tested; starting exactly at the equilibrium would give a constant solution.

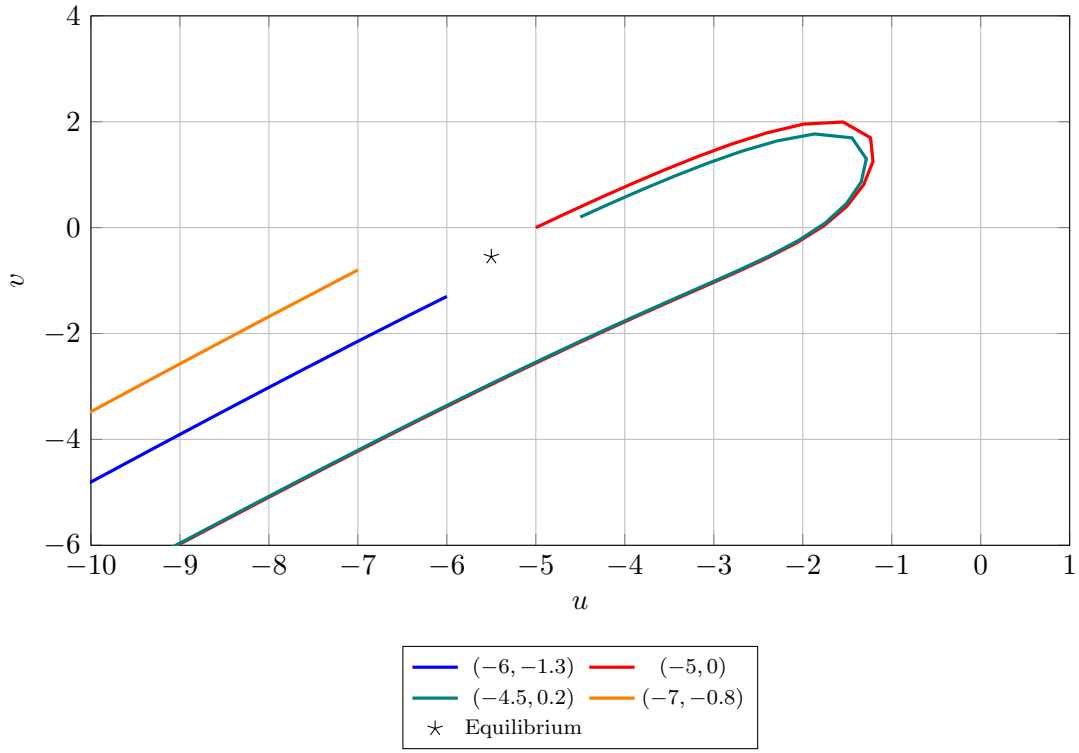


Figure 2: Four trajectories near the merged equilibrium at $G = -3.025$.

Case 3: $G = 1.9 > G_{\mathrm{SN}}$

For the final case, the quadratic formula gives two equilibria:

$$(u_1^*, v_1^*) \approx (1.518, 0.152), \qquad (u_2^*, v_2^*) \approx (-12.518, -1.252).$$

The Jacobian eigenvalues at the first point are approximately -0.419 and -3.351, so it is a **stable node**. At the second point, the eigenvalues are approximately -0.119 and 11.789, so it is a **saddle point**.

I tested six starting points to see whether the stable node would attract all of them:

Initial condition (u_0, v_0)	**Observed result**
$(-8, -1.7)$	Uncontrolled
$(-6, -0.4)$	Controlled
$(-7, -1.3)$	Controlled
$(-4, -1.4)$	Controlled
$(-5, -0.6)$	Controlled
$(-9, -1.9)$	Uncontrolled

Four trajectories approached the stable node, while two escaped. Therefore, even when a stable equilibrium exists, control still depends on the initial condition.

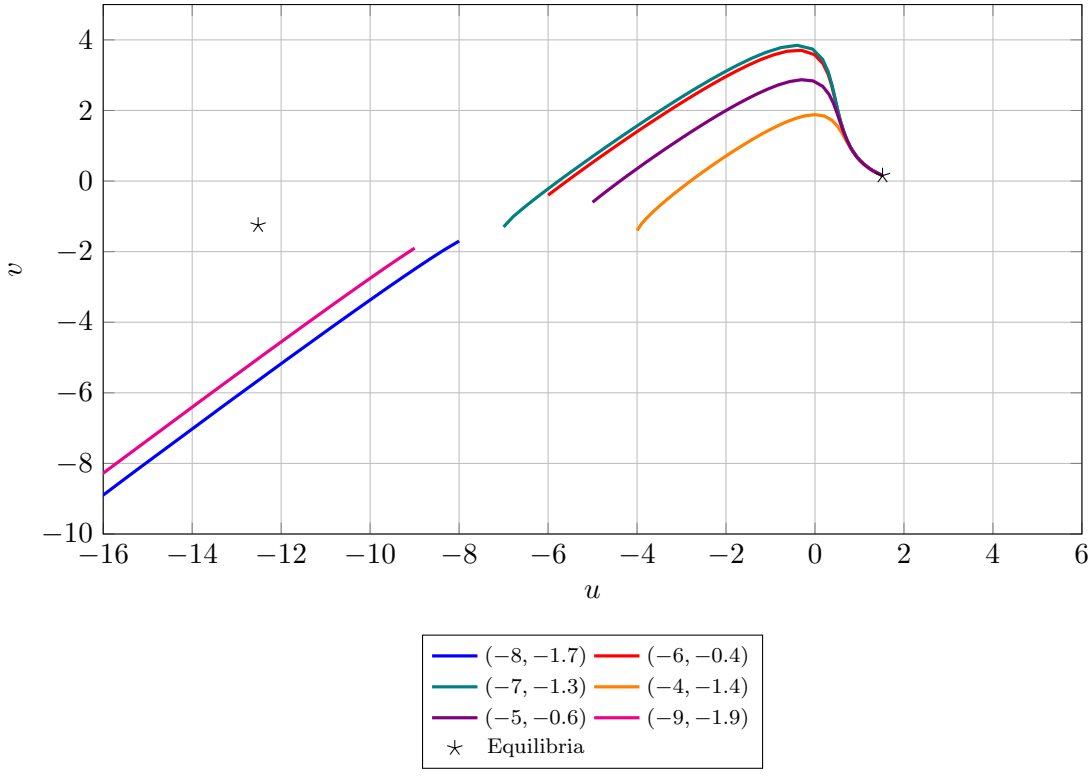


Figure 3: The six tested trajectories for $G = 1.9$, including both outcomes.

Conclusions on Controllability

Based on the simulations and the equilibrium calculations, I can make the following statements for the parameters tested:

- **For $a = 0.1$, $b = 1.0$, and $G < -3.025$, the model cannot settle at a finite equilibrium.** There are no real steady states in this range, and all four trajectories I simulated at $G = -3.7$ were uncontrolled.
- **For $a = 0.1$, $b = 1.0$, $c = 1.0$, and $G = -3.025$, none of my four tested initial conditions were controlled.** The single equilibrium $(-5.5, -0.55)$ has an unstable direction, although this does not rule out special initial conditions that remain at or approach it.
- **For $a = 0.1$, $b = 1.0$, $c = 1.0$, and $G = 1.9$, the behaviour is controlled for the tested initial conditions** $(-6, -0.4)$, $(-7, -1.3)$, $(-4, -1.4)$, **and** $(-5, -0.6)$**.** These trajectories approach the stable node near $(1.518, 0.152)$. However, the tested points $(-8, -1.7)$ and $(-9, -1.9)$ were uncontrolled.

These results suggest that $G > G_{\mathrm{SN}}$ makes controlled behaviour possible, but it does not guarantee control from every starting point. The initial condition must also lie in the region attracted to the stable equilibrium.